

Learning to Detect Industrial Time-Series Anomalies From Imputation Consistency With Sparse Observations

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Abstract—Time-series anomaly detection plays an important role in ensuring industrial safety. Currently, many anomaly detection methods mainly target complete time series and ignore the widespread problem of data missing in the real world. Therefore, this article proposes a novel anomaly detection method for time series with sparse observations based on imputation consistency using a mixture of patch information inference network (MoPIN). Due to the robustness of the imputation method modeling to the random mask, different imputed series of the same normal time series with different random masks should have consistency. Then, a novel imputation consistency is used to detect anomalies in sparse observation series. Moreover, the MoPIN imputes series by a two-step imputation and multiscale modeling of patch information. Meanwhile, the similarity of imputed series under different masks is used to measure imputation consistency, which well constructs the relationship between sparse observation series and anomaly scores. Finally, the MoPIN can accurately detect anomalies while imputing series. Extensive experiments on four real-world benchmarks in different domains of imputation and anomaly detection tasks and a real fluid catalytic cracking (FCC) process case demonstrate the effectiveness of the proposed method. Specifically, the MoPIN achieved at least 8.05% mean absolute error (MAE) relative improvement in imputation and 3.74% F1 relative improvement in anomaly detection.

Index Terms—Anomaly detection, patch, process industry, sparse observation, time series.

NOMENCLATURE

Abbreviation	Expansion
X, T , and C	Time series and its length and dimension.
X^o and X^m	Sets of observed values and missing values of X .
Δ, O , and L	Sampling interval, observation, and delay matrix, respectively.
l, s , and n	Length, step size, and the number of patches, respectively.
S, S_t , and S_s	Input series of the PIML and its trend and seasonal components, respectively.

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W and b	Weight matrix and bias vector, respectively.
d and h	Embedding and encoder dimension, respectively.
$\check{H}, \hat{H}, \acute{H}$, and $\grave{H}$	Input of encoders.
$\check{Z}, \hat{Z}, \acute{Z}$, and $\grave{Z}$	Output of encoders.
$\tilde{Z}$	Spatiotemporal mixture feature.
U and V	Output of PIML and PIMM, respectively.
θ	All parameters of the MoPIN.
R, X^R, O^R , and L^R	Random mask matrix and time series and its corresponding observation and delay matrix.
C_I and λ	Imputation consistency constraint and coefficient.
X^{Imp} and X^{Rec}	Imputation and reconstruction series, respectively.
L_{Stage1} and L_{Stage2}	Optimization target of two stages.

I. INTRODUCTION

WITH the development of industrial technologies, the number of industrial devices is increasing. The occurrence of device anomalies poses a serious threat to industrial safety [1]. Accurate status monitoring of these devices plays an important role in ensuring industrial production safety and improving production efficiency [2]. However, with a large number of devices, it is difficult to monitor manually. Therefore, it is of practical significance to achieve accurate and timely status monitoring through anomaly detection algorithms. Time series exist in various industrial scenarios and are widely used in industrial anomaly detection, including process industries, spacecrafts, and industrial servers. However, due to the rarity of actual anomalies and the professional nature of identification, labeling data are time-consuming and difficult. Therefore, unsupervised time-series anomaly detection is an active research topic [3].

Currently, promising progress has been made in unsupervised time-series anomaly detection, including traditional methods and deep learning-based methods [4]. Traditional methods identify anomalies through spatial projection and iterative partitioning. Deep learning-based methods use their powerful feature extraction capabilities to distinguish anomalies [5]. However, traditional methods ignore the inherent time dependencies, making it difficult to achieve accurate anomaly detection. Therefore, deep learning-based methods

have been widely studied, which are divided into prediction-based methods and reconstruction-based methods [1]. The anomaly data are difficult to accurately predict or reconstruct by the model modeled by normal data, so time-series anomaly detection methods based on recurrent neural network (RNN) [6], graph neural network (GNN) [1], Transformer [7], and many other basic neural networks have been proposed to realize data representation and anomaly discovery through feature extraction of different dimensions and capabilities.

However, current anomaly detection methods are all used for complete time series, but in real industrial scenarios, data missing due to reasons such as communication interruption and nonuniform sampling is very common [8]. At present, anomaly detection methods for time series with sparse observations often adopt a two-stage approach, which use imputation methods to impute series and then use anomaly detection methods to detect anomalies [9]. Obviously, the above approach breaks the connection between the data and the target of anomaly detection, hindering to accurately identify anomalies.

Benefiting from the advantages of deep learning methods in deep modeling of time series, deep learning-based time-series imputation methods have also achieved remarkable results [10]. Therefore, deep learning methods are used to jointly perform two tasks of imputation and anomaly detection to achieve single-stage anomaly detection for sparse observation series. Time-series imputation methods use random masking to train models since the imputation model should be robust to random masks. Moreover, due to factors such as the inherent time dependence and dimensional dependence in time series, different random mask positions in a series will not cause significant changes in its imputed series. Combined with the idea of unsupervised time-series anomaly detection, when only normal samples are used for modeling, anomaly series are difficult to be accurately reconstructed or predicted. Meanwhile, for normal time series, the imputed time series with different masks should have consistency. And when the series contains anomalies, since the model itself lacks the ability to recover the anomaly data, it is difficult to achieve robust imputation under different masks. Therefore, series with anomalies will have lower imputation consistency than normal series, which can be used to combine imputation and anomaly detection tasks to achieve anomaly detection of time series with sparse observations.

In addition, patch-based approach is used to implement series modeling, which is mainly due to its powerful local-global modeling capabilities [11]. In the field of time-series analysis, the patch-based approach provides the time-series model with partial frequency domain information at the input level, enabling it to achieve better information extraction and utilization [12]. Most recent patch-based works use the channel-independence approach for series modeling, achieving better results than channel-mixture modeling. However, this approach ignores the feature relationship between channels [13]. Therefore, this article will adopt the hybrid modeling approach for patch information, integrating the advantages of channel-independence modeling and mixture spatiotemporal modeling.

In summary, this article introduces a patch-based approach to time-series anomaly detection with sparse observations and proposes a mixture of patch information inference network (MoPIN) for imputation consistency learning. Specifically, a hybrid modeling approach based on channel independence and mixture for patch information is proposed. In addition, a two-step time-series imputation framework is designed to achieve accurate imputation. Furthermore, the imputation consistency of different mask series is proposed and exploited for detecting anomalies. In particular, despite the complex design, all branches of the MoPIN are implemented by simple multilayer perceptions (MLPs). Finally, the MoPIN can fully integrate the tasks of time-series imputation and anomaly detection, and thus realize anomaly monitoring of time series with sparse observations while performing imputation. The contributions are summarized as follows.

- 1) This article proposes a novel MoPIN to realize time-series modeling by making full use of the information of time-series patches.
- 2) The MoPIN not only accurately imputes series, but also uses the imputation consistency to achieve anomaly detection of time series with sparse observations, thus avoiding the separation between data and tasks.
- 3) Extensive experiments on four real-world benchmarks in different fields of series imputation and anomaly detection and a real chemical production application case demonstrate the effectiveness of the proposed method. Specifically, the MoPIN achieved at least 8.05% mean square error (mse) relative improvement in imputation and 3.74% *F1* relative improvement in anomaly detection.

The rest of this article is organized as follows. Section II reviews the related work. Section III introduces the proposed MoPIN method. Section IV analyzes the performance of the MoPIN through extensive experiments. Section V verifies the application effect of the MoPIN using an actual industrial case. Section VI draws the conclusions of this article.

II. RELATED WORKS

This section reviews recent related works of time-series anomaly detection and imputation. And also, the patch-based time-series modeling methods are reviewed.

A. Unsupervised Time-Series Anomaly Detection

Due to the scarcity of data labels, traditional unsupervised methods often score data points at a timestamp based on density or distance. The typical method, principal component analysis (PCA) [14], achieves anomaly identification by projecting data points into a low-dimensional space and using statistics as its distance measure [15]. The isolation forest (IF) [5] divided the data space through iterative partitioning and calculated the local density of data points to identify anomalies. The recent copula-based outlier detection (COPOD) method [16] and the empirical cumulative distribution-based outlier detection (ECOD) method [17] determined the anomalies by calculating the probability density of data points. However, although traditional methods can achieve data anomaly detection faster, they only consider the information of

data points at a single timestamp and ignore the inherent time-dependent characteristics, which hinders accurate anomaly detection.

Benefiting from the powerful feature extraction capability of deep learning, Zong et al. [18] combined the traditional density-based method with the deep model and proposed the deep autoencoder Gaussian mixture model (DAGMM). However, it still does not consider the data characteristics in the time dimension. Recent deep learning-based methods further take the characteristics of the time dimension into consideration and achieve anomaly detection through series prediction or reconstruction errors. In terms of prediction-based methods, Hundman et al. [6] used a long-short term memory (LSTM) network to build a time-series model and detected anomalies through prediction errors. Chen et al. [3] and Zhang et al. [1] used the GNN to construct the graph structure with a Transformer for anomaly detection (GTA) and the graph structure change-based anomaly detection method (GSC-MAD), and also used prediction errors to detect and locate anomalies. Meanwhile, reconstruction-based methods have also been fully developed. Based on the widely used autoencoder, Audibert et al. [19] and Tuli et al. [7] adopted adversarial training to construct the unsupervised anomaly detection (USAD) method and the Transformer-based anomaly detection (TranAD) method, respectively. Wu et al. [20] transformed the series into two dimensions and used a typical network framework for processing 2-D data to extract the frequency-periodic features of the data, reconstructed the series, and used the error to detect anomalies. In addition, Han et al. [21] proposed a self-supervised memory enhanced deep clustering (SMEL) method combining temporal dimension information autoencoder and clustering idea to detect anomalies in an actual wastewater treatment process. Zhou et al. [22] also detected anomalies using clustering idea and temporal information extraction based on dynamic graph and entity-aware normalizing technologies.

However, the above methods are all studied for complete series. For series with missing values, these methods need to fill in the series before detection, which makes it impossible to directly connect the data and the task, and the performance of anomaly detection is directly affected by the performance of imputation, which hinders the performance of anomaly detection [9]. Recently, Xiao and Fan [23] integrated imputation and anomaly detection process using an end-to-end framework (ImAD) to impute incomplete anomaly data and detect anomalies based on well-designed pseudo-abnormal samples. Different from the above method, this article is characterized by a novel imputation consistency-based anomaly detection condition, which jointly designs the time-series model for anomaly detection and imputation tasks to learn a direct connection between data and anomaly scores.

B. Time-Series Imputation

In real-world scenarios, data missing is a common problem and can be solved by the imputation method, which is divided into traditional methods and deep learning-based methods [24]. Traditional methods impute missing values by copying the observation around it or using statistical indicators

of series [25]. The classic mean imputation method uses the mean of the series to fill in the missing values. In addition, some works adopt the iterative calculation approaches including soft threshold singular value decomposition, matrix decomposition, and random forest to propose imputation methods such as soft imputation (SoftImpute) [26], matrix factorization imputation (MatrixFactorization) [27], and missing forest (MissForest) [28]. However, the above methods have the problem of strong assumptions about missing values, which introduces strong biases in imputed series.

Recently, deep learning-based imputation methods have achieved promising progress, which automatically mines the inherent information in series through deep representation and data recovery, without considering the strong assumptions required by traditional methods. Some researchers used bidirectional RNN and hidden layer state propagation to propose a bidirectional recurrent imputation method for time series (BRITS) [29] and multidirectional RNN (M-RNN) [30] to fill in missing values, respectively. Marisca et al. [31] proposed a novel GNN-based method with the spatiotemporal point inference network (SPIN), which imputed missing values by learning the spatiotemporal graph representation of sparse data. In addition, the TimesNet [20] was a general time-series analysis tool that used convolutional neural networks to extract the period-frequency information of series and impute in missing values. Du et al. [10] adopted the diagonal mask self-attention mechanism and combined it with weighted combination design to propose a two-step self-attention mechanism-based imputation method for time series (SAITS). Yuan et al. [8] proposed the attention-based interval aided network (AIA-Net) combining the self-attention mechanism with the RNN, which was used for soft measurement of a hydrocracking process. Pan et al. [32] combined data statistics and autoencoders to design an adaptive learning strategy for proposing the adaptive-learned median-filled deep autoencoder (AM-DAE) in the soft sensor of a hydrocracking device.

The above methods are all designed for imputation, which is subject to achieve better downstream tasks such as classification and prediction. However, as aforementioned, this two-step approach leads to a gap between the data and downstream tasks, making the performance of downstream tasks directly affected by imputation. In order to improve the performance of downstream tasks, Che et al. [33] used gated recurrent units to extract features of series with missing values, while allowing the network to directly participate in the calculation of classification task, thereby improving the accuracy of classification tasks. Miao et al. [24] introduced additional classifiers in the generative adversarial network (GAN) used for data imputation, and improved the classification performance of time series with missing values through semi-supervised learning. Different from the above methods, this article uses inherent imputation consistency in the time-series imputation task to detect anomalies and closely links the relationship between data and anomaly detection task in the joint optimization of the two tasks.

C. Patch-Based Time-Series Method

Recently, patch-based methods have shown great capabilities in the field of time-series analysis. Motivated by

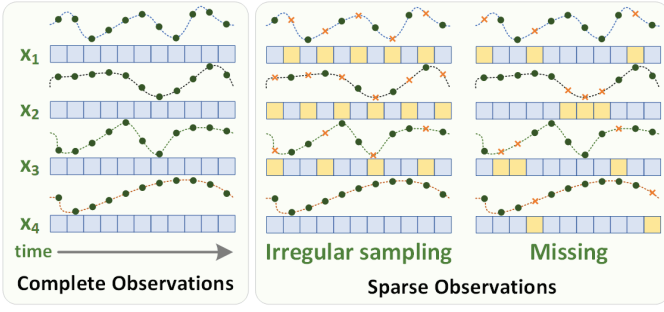


Fig. 1. Time series with complete and sparse observations.

natural language processing, Nie et al. [11] divided time series into patches, arguing that each patch could be treated as a word, and proposed the patch time-series Transformer method (PatchTST) for long-term time-series forecasting. Subsequently, a wide range of patch-based methods were proposed. Chen et al. [12] extracted interpatch and intrapatch features based on patch segmentation, combined temporal resolution and distance information for multiscale modeling, and proposed multiscale Transformers with adaptive pathway (Pathformer) for long-term series forecasting. In time-series anomaly detection, Yang et al. [4] proposed a dual-branch attention structure based on patchwise and in-patch to learn the permutation invariant representation of series, and learned the dual-branch difference representation under the constraint of contrastive loss to perform anomaly detection. However, under the advantage of channel independence modeling, the above methods ignore the relationship between series channels. Unlike the previous usage of patches, the proposed MoPIN utilizes different advantages of channel independence modeling and spatiotemporal mixture modeling after time-series patching, and constructs a time-series model by independent modeling and result fusion to achieve full utilization of information.

III. METHOD

For ease of understanding, this section describes the problem definition. Then, the patch information mixture module (PIMM), which is the core module of the MoPIN is introduced. Furthermore, the overall design of the MoPIN is described. Finally, imputation consistency and optimization objectives are detailed.

A. Problem Definition

For a system with C variables uniformly sampled over time, the observation series $X \in \mathbb{R}^{T \times C}$ over a time period of length T is a full observation time series. If values of a variable are not observed within time T , the observation series $X = \{X^o, X^m\}$ is a time series with sparse observations, where X^o and X^m denote the sets of observed values and missing values, respectively. Fig. 1 shows an illustration of a time series of complete and sparse observations containing four variables.

In the modeling of time series with sparse observations, two auxiliary matrices are introduced to utilize the information contained in the data. The first is the observation matrix $O \in \mathbb{R}^{T \times C}$, which indicates whether the observation is missing at a timestamp t , and the second is the delay matrix $L \in \mathbb{R}^{T \times C}$,

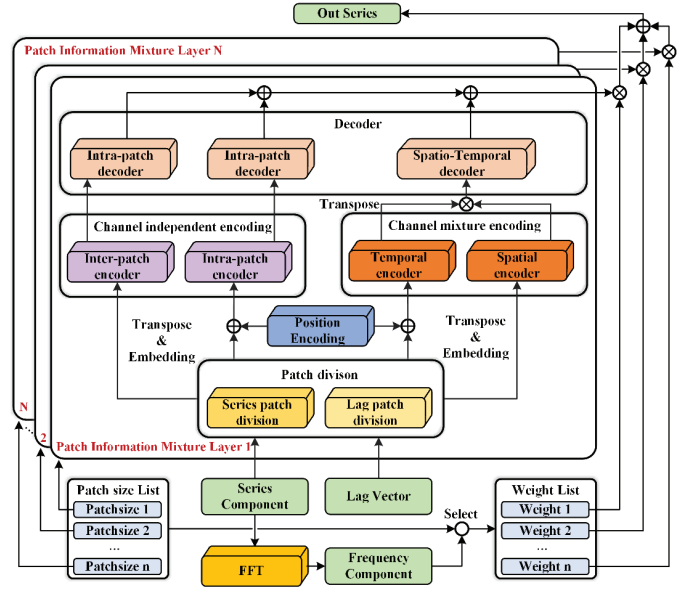


Fig. 2. Architecture of the PIMM.

which indicates the time interval between the time of the latest historical observation and the current time t . Suppose sampling interval is Δ , matrices O and L can be calculated by the following equations:

$$O_{t,c} = \begin{cases} 1, & \text{if } x_{t,c} \text{ is not missing} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$L_{t,c} = \begin{cases} 0, & \text{if } t = 1 \\ \Delta, & \text{if } t > 1 \text{ and } O_{t-1,c} = 1 \\ L_{t-1,c} + \Delta, & \text{if } t > 1 \text{ and } O_{t-1,c} = 0. \end{cases} \quad (2)$$

And the USAD for time series with sparse observations is to use $\{X, O, L\}$ to determine whether the system status at timestamp t is anomalous or not without labels.

B. Patch Information Mixture Module

1) *Multiscale Modeling*: The effectiveness of multiscale modeling in time-series analysis has been widely verified [20]. Therefore, multiscale modeling is introduced based on the patch size in the PIMM, as shown in Fig. 2, which consists of N independent patch information mixture layers (PIMLs), each of which uses patch information of different scales. Finally, the output of the PIMM is obtained by weighting the calculation results of each layer.

Specifically, given the patch size l and the step size s , a time series can be divided into different overlapping or nonoverlapping patch series, as shown in Fig. 3. In this way, the information about different scales of the original series can be extracted from different patch series. In particular, after multiscale patch division, the positional relationship between missing values and observations in the series can be well-connected. Therefore, the MoPIN can impute missing values well through utilizing local information around them.

In each PIML, patch division is performed on the input series S and the corresponding delay matrix L . The dimension size of S and L after division is $\mathbb{R}^{l \times n \times C}$, where $n = \lceil (T - l)/s \rceil + 1$ denotes the number of the patch in each channel.

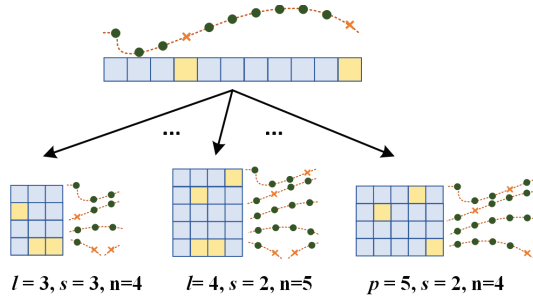


Fig. 3. Multiscale patch division.

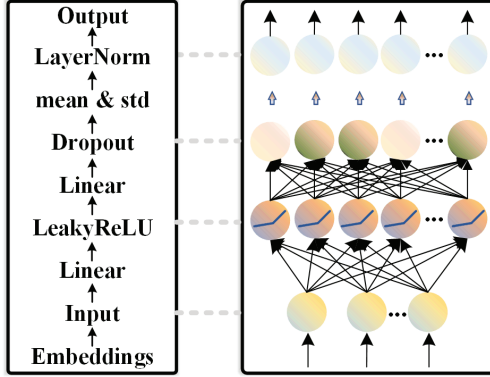


Fig. 4. MLP encoder.

Then, the PIML encodes S and L by channel-independence encoding and channel-mixture encoding, respectively.

2) *Channel Independence and Mixture Encoding*: In previous works, both channel-independence modeling [11] and channel-mixture spatiotemporal modeling [1] were found to be beneficial for time-series analysis. And some proposed serial modeling approaches [13] using both of above approaches may lead to information leakage between channels during channel-wise modeling. Therefore, the MoPIN blends the advantages of the two modeling approaches in a parallel manner. In addition, the MoPIN directly decodes the results encoded by each encoder and then fuses them, so the MoPIN can fully utilize the advantages of both modeling approaches. Specifically:

- 1) *Channel independence encoding*. This encoding module encodes the information within the patch (intrapatch) and between patches (interpatch) separately. Among them, the interpatch encoder encodes the global granularity information between different patches, and the intrapatch encoder encodes the temporal fine-grained information between different timestamps in the patch of a single channel.
- 2) *Channel mixture encoding*. This encoding module encodes the temporal and spatial information of the patch separately and fuses them. In detail, the temporal and spatial encoders encode the relationships between the time and channel dimensions of the patch, separately.

All of the four encoders are achieved by the MLPs with the same structure as shown in Fig. 4 and calculated in different dimensions to extract the multiscale information, which can be calculated by the following equation:

$$Z = \text{LayerNorm}(\text{Dropout}(\text{LeakyReLU}(H \times W_1 + b_1) \times W_2 + b_2)) \quad (3)$$

where $W_1 \in \mathbb{R}^{d \times h}$ and $W_2 \in \mathbb{R}^{h \times d}$ are the weights, $b_1 \in \mathbb{R}^{h \times 1}$ and $b_2 \in \mathbb{R}^{d \times 1}$ are the bias terms, and H and Z with the same dimension denote the input and output of the encoder, respectively. And d and h denote the embedding dimension and the hidden layer dimension of the MLP, respectively. By transforming the dimension of the input series S , the interpatch, intrapatch, temporal, and spatial encoders can extract the corresponding information $Z = \{\tilde{Z}, \hat{Z}, \check{Z}, \bar{Z}\}$ from the embedding $H = \{\tilde{H}, \hat{H}, \check{H}, \bar{H}\}$ with dimensions of $\mathbb{R}^{C \times l \times d}$, $\mathbb{R}^{C \times n \times d}$, $\mathbb{R}^{n \times l \times d}$, and $\mathbb{R}^{n \times C \times d}$. In addition, for the spatiotemporal mixture encoding, spatiotemporal mixture feature $\tilde{Z} \in \mathbb{R}^{C \times n \times l}$ can be calculated by the following equation:

$$\tilde{Z} = \text{Transpose}(\tilde{Z} \times \tilde{Z}^T) \quad (4)$$

where $\text{Transpose}(\cdot)$ is the dimension permutation operation. Then, $\tilde{Z}$, $\hat{Z}$, and $\bar{Z}$ are decoded and the output of the n th PIML $U_n \in \mathbb{R}^{C \times T}$ can be calculated by the following equation:

$$U_n = \tilde{Z} \times \tilde{W}_D + \hat{Z} \times \hat{W}_D + \bar{Z} \times \bar{W}_D \quad (5)$$

where $\tilde{W}_D \in \mathbb{R}^{l \times d \times T}$, $\hat{W}_D \in \mathbb{R}^{n \times d \times T}$, and $\bar{W}_D \in \mathbb{R}^{n \times l \times T}$ are the weights of the decoder.

3) *Embedding*: For different encoders, the inputs with different dimensions are embedded into the same dimension for calculation via linear projection. Meanwhile, the delay matrix as auxiliary information is embedded to assist imputation. Different from previous methods that also use delay information, the MoPIN embeds it through learnable parameters rather than using it by concatenation. In this way, the delay matrix can make the model pay attention to its neighboring local observations during imputation and avoid mutations caused by concatenation. In addition, for the two encoders that extract temporal granularity information, the positional encoding of the series is also embedded. Therefore, in addition to the positional encoding, the embedding H can be calculated by the following equation:

$$H = \text{Transpose}(S) \times W_S + \text{Transpose}(L) \times W_L \quad (6)$$

where W_S and W_L are the weights of the embedding calculation.

4) *Adaptive Aggregate*: Motivated by the TimesNet [19], it can be believed that patches of different scales can better extract different frequency-period features of the series. Therefore, the frequency amplitude calculated by the fast Fourier transform is used as the weight of the scale corresponding to different patch sizes, and performs adaptive aggregation on the output of the PIML. For N PIMLs, the patch size set $P = \{l_1, l_2, \dots, l_N\}$ and the output set of PIMLs $U = \{U_1, U_2, \dots, U_N\}$, the output of the PIMM V can be calculated by the following equations:

$$A = |\text{FFT}(S)| \quad (7)$$

$$W_A = \{w_A^1, w_A^2, \dots, w_A^N\} = A_P \quad (8)$$

$$V = U \times W_A \quad (9)$$

where $\text{FFT}(\cdot)$ denotes the fast Fourier transform and $W_A \in \mathbb{R}^{N \times 1}$ is the weight vector.

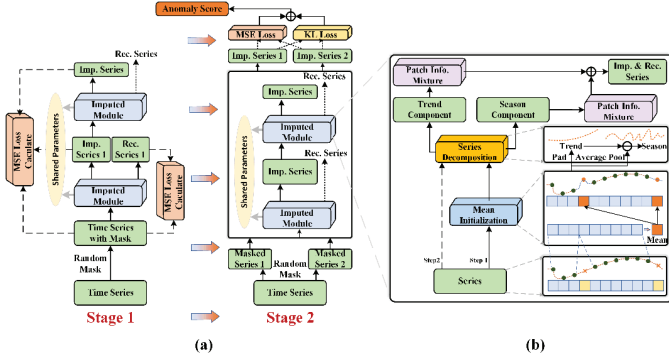


Fig. 5. (a) Overview of the MoPIN. (b) Structure of the imputed module.

C. MoPIN

Fig. 5(a) shows the overall structure of the MoPIN. Overall, the MoPIN achieves anomaly detection while imputing time series. Different from the current two-stage anomaly detection method for time series with sparse observations, the two stages of the MoPIN are the two stages of model construction. In the first stage, the pretrained model $F(X; \theta_{\text{pre}})$ with parameters θ_{pre} is obtained through a two-step imputation only with the constraint of the imputation task. Specifically, the second-step imputation is a modification of the first-step imputation result by the same imputation module with shared parameters. In the second stage, the imputation consistency is used to constrain the model and the model $F(X; \theta_{\text{tune}})$ with updated parameters θ_{tune} is trained, which is suitable for the anomaly detection task. Through the two-stage design, the MoPIN closely links the imputation and anomaly detection tasks, while directly calculate anomaly scores during the imputation process without the need for additional anomaly detection methods.

In both stages, the MoPIN performs a random mask operation on time series to facilitate the training of the imputation and the calculation of imputation consistency. The random mask operation can be calculated as

$$X^R = O \odot (1 - R) \odot X \quad (10)$$

where $R \in \mathbb{R}^{T \times C} = \{0, 1\}$ denotes the random mask matrix, in which 1 means the value is masked. And $X^R = \{X^{R^m}, X^{R^o}\}$ is the masked series and the corresponding observation matrix O^R and delay matrix L^R are calculated by (1) and (2), respectively. And the $\odot$ is the elementwise multiply. The mask ratio is $r = (||R||_1 / (T \times C))$.

At any stage, the imputed module contains all the forward propagation calculations, and its overall structure is shown in Fig. 5(b). It decomposes the series into trend and seasonal components through a series decomposition module and uses, respectively, two different PIMMs to build decomposed series models, and then merged their results to achieve imputation and reconstruction. Here, imputation and reconstruction correspond to missing values X^{R^m} and observed values X^{R^o} , respectively. In addition, in the first step, the mean of the series is used to initialize X^{R^o} , which can be calculated by

the following equation:

$$X^{R^o}_{i,j} = \frac{\sum X^{R^o}_{:,j}}{\sum O^R_{:,j}}. \quad (11)$$

Compared with zero-imputation, this can better utilize the statistical information in series and avoid meaningless mutations caused by zero-imputation, which is very important for anomaly detection tasks. In addition, the series decomposition module uses global average pooling to extract the trend component, and the rest is considered as the seasonal component, which can be calculated by the following equations:

$$S_t = \text{AvgPool}(\text{Pad}(S)) \quad (12)$$

$$S_s = S - S_t \quad (13)$$

where S , S_t , and S_s denote the input series, trend, and seasonal component, respectively. The $\text{AvgPool}(\cdot)$ and $\text{Pad}(\cdot)$ are the global average pooling and padding operations, respectively.

D. Imputation Consistency

As analyzed in the Introduction, the training of imputation models is often achieved through the recovery of random masked values. For a robust imputation model, different random masks usually do not cause huge changes in imputed values, which is also the premise of imputation model training. However, referring to the idea of unsupervised time-series anomaly detection, intuitively, when the series contains anomalies, different random masks will lead to large differences between the imputed series. This is because anomalies destroy the robustness of the proposed model. Therefore, the MoPIN exploits this to detect anomalies.

Specifically, the MoPIN measures the imputation consistency of two imputed series by the mse and distribution similarity between them. On the one hand, the mse is an intuitive measure of the consistency of the imputation results. On the other hand, according to the work of Miao et al. [24], even with a small mse, the distribution of the imputed series may be significantly different from the original series. Therefore, the distributional similarity of the two imputed series is incorporated as a constraint. Assume that the two random mask series of the time series X are $\tilde{X}$ and $\check{X}$, and the corresponding imputed series are $\tilde{X}^{\text{Imp}}$ and $\check{X}^{\text{Imp}}$. Then, the imputation consistency C_I is defined as follows:

$$C_I = (1 - \lambda) \|\tilde{X}^{\text{Imp}} - \check{X}^{\text{Imp}}\|_2^2 + \lambda \frac{1}{2} (D_{\text{KL}}(\tilde{X}^{\text{Imp}} \parallel \check{X}^{\text{Imp}}) + D_{\text{KL}}(\check{X}^{\text{Imp}} \parallel \tilde{X}^{\text{Imp}})) \quad (14)$$

where $D_{\text{KL}}(p \parallel q) = -\sum p \ln(p/q)$ denotes the Kullback-Leibler divergence, which is used to measure the distribution similarity of two series. And λ denotes the balance hyperparameter. The MoPIN uses the calculated imputation consistency as the anomaly score and compares it with a predefined threshold for anomaly detection.

E. Optimization

According to different objectives of the two stages of the MoPIN, different optimization functions are designed.

Stage 1: The goal of the first stage is to enable the MoPIN to effectively impute missing values. Therefore, according to

TABLE I
DETAILS OF BENCHMARKS

Tasks	Datasets	Channels	Samples	Missing rate	Information	Anomaly ratio
Imputation	ETTh1	7	8545/2785/3505	0%	Electricity	-
	Air	11	219756/72492/125100	1.60%	Air-Quality	-
Anomalydetection	PSM	25	99289/33097/87746	3.17	Service	0.278
	MSL	55	4031/249/16502	0%	Spacecraft	0.115
	SMAP	25	501/168/1956			0.167

the two-step imputation process, the following optimization function is designed. Suppose that the output of the imputation module is Y , then for observed values and missing values, the imputed series X^{Imp} and the reconstructed sequence X^{Rec} can be obtained, respectively,

$$X^{\text{Imp}} = Y \odot (1 - O^R) + X^R \odot O^R \quad (15)$$

$$X^{\text{Rec}} = Y \odot O^R. \quad (16)$$

Then, for the first step, the MoPIN uses L2 loss to optimize both reconstruction and imputation tasks, and its optimization target L_{Stage1}^1 is shown in the following equation:

$$L_{\text{Stage1}}^1 = \left\| (X_1^{\text{Imp}} - X) \odot R \right\|_2^2 + \|X_1^{\text{Rec}} - X^R\|_2^2 \quad (17)$$

where X_1^{Imp} and X_1^{Rec} are the imputed and reconstructed series for the first step. Then, for the second step, only the imputed series is constrained, and the optimization target L_{Stage1}^2 is shown in the following equation:

$$L_{\text{Stage1}}^2 = \left\| (X_2^{\text{Imp}} - X) \odot R \right\|_2^2 \quad (18)$$

where X_2^{Imp} is the second imputed series. Hence, the optimization target of the first stage is shown in the following equation:

$$L_{\text{Stage1}} = L_{\text{Stage1}}^1 + L_{\text{Stage1}}^2. \quad (19)$$

Stage 2: For the second stage, the imputation consistency is directly used as the optimization objective L_{Stage2}

$$L_{\text{Stage2}} = C_I. \quad (20)$$

Therefore, the MoPIN can achieve two tasks: imputation and anomaly detection. For the imputation, the MoPIN is optimized through the L_{Stage1} . For the anomaly detection, the MoPIN uses L_{Stage1} as the target and optimizes it a certain number of times, and then uses L_{Stage2} to continue optimizing and the imputation consistency is used as the anomaly score to detect anomalies. In general, the two stages of the MoPIN can be expressed as

$$X^{\text{Imp}} = F(X; \theta_{\text{pre}}), \quad \text{s.t. } L_{\text{Stage1}} \quad (21)$$

$$\text{AnomalyScore} = F(X; \theta_{\text{tune}}), \quad \text{s.t. } L_{\text{Stage2}}. \quad (22)$$

IV. EXPERIMENT

This section verifies the imputation and anomaly detection performance of the MoPIN through multiple real-world benchmarks and illustrates its effectiveness through comparative experiments, ablation, and efficiency analysis.

A. Benchmarks and Evaluation Metrics

1) *Benchmark:* The MoPIN is validated on four real-world benchmarks of two typical time-series tasks, and the details are shown in Table I: 1) electricity transformer temperature (ETT); 2) Beijing Multi-Site Air-Quality (Air); 3) Mars Science Laboratory (MSL) and Soil Moisture Active Passive (SMAP); and 4) Pooled Server Metrics (PSM). Specifically, in order to verify the performance of the MoPIN on time series with sparse observations, 10% of the observations in the benchmarks are randomly masked. And on the validation set and test set, these 10% mask observations are used as ground truth to evaluate. In addition, for the MSL and the SMAP [6], only three nontrivial subsets A-4, C-2, and T-1 are used because the rest of the subsets are mostly trivial series [7]. Furthermore, for fairness in evaluation, the length of all series is set to 96, and all benchmarks are preprocessed by standardization. And following the protocol of previous works [10], the benchmarks are split into training, validation, and test sets chronologically by the ratio of 6:2:2 for the ETTh1 and 7:1:2 for the Air. For the anomaly detection task, the training data are split into training and validation sets according to the ratio of 3:1 [1].

2) *Evaluation Metrics:* The standard metrics are utilized for evaluation. For imputation task, the mse ($\text{mse} = (1/\|R\|_1)\|(X^{\text{Imp}} - X) \odot R\|_2^2$), the mean absolute error ($\text{MAE} = (1/\|R\|_1)\|(X^{\text{Imp}} - X) \odot R\|_1$), and the root-mean-square error ($\text{RMSE} = \sqrt{\text{MSE}}$) are used. For anomaly detection, precision [$P = (\text{TP}/(\text{TP} + \text{FP}))$], recall [$R = (\text{TP}/(\text{TP} + \text{FN}))$], and $F1$ score [$F1 = (2 \times P \times R)/(P + R)$] are used, where TP and FP denote the correctly and incorrectly detected anomalies, respectively, and TN and FN denote the correctly and incorrectly labeled normal, respectively. Specifically, due to different prediction or reconstruction performances between different anomaly detection methods, the previous evaluation protocol is used, which is that the best thresholds for all methods are selected through grid search, and the best performances for them are reported for all experiments [3]. Moreover, the point adjust strategy is used [34].

B. Baselines

For the imputation task, the MoPIN is compared with 12 baselines, including four traditional methods with the Mean, the MissForest [28], the SoftImpute [26], and the MatrixFactorization [27], and eight latest deep learning-based methods with the AMDAE [32], the BRITS [29], the MRNN [30], the AIA-GRU [8], the SPIN [31], the TimesNet [20], the PatchTST [11], and the SAITS [10]. Among them, the TimesNet is the current state-of-the-art (SOTA) method.

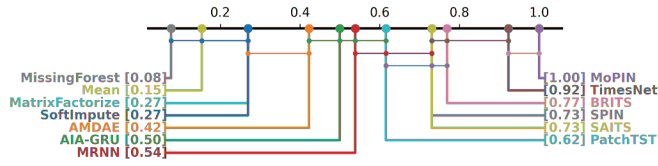


Fig. 6. Critical difference diagram of the metric 1/mse.

For the anomaly detection task, the MoPIN is compared with ten baselines, including traditional methods with the KPCA [14], the IF [5], the COPOD [16], the ECOD [17], and deep learning-based methods with the DAGMM [18], the OmniAnomaly [34], the USAD [19], the CAE_M [35], the DCDetector [4], the TimesNet [20], and the ImAD [23]. Among them, the DCDetector and the ImAD are the current SOTA methods for time series without and with missing values. In particular, all the baselines except the ImAD use the TimesNet to impute missing values before detect anomalies.

C. Implementation Details

We implement the MoPIN, its variants, and all baselines in PyTorch version 1.8.2 with CUDA 10.2. All experiments are conducted on a server with Intel¹ Xeon² Silver 4210 CPU at 2.20 GHz and a single NVIDIA TITAN V 12-GB GPU. For fairness, the consistent hyperparameters are used for all experiments. The random mask ratio r is set to 0.1, and the number of the PIMs N is set to 2. Hence, patch sizes are set to {16, 24}, and related strides are set to {8, 12}. Moreover, the embedding and encoder dimensions d and h are set to 16 and 8, respectively. And for the imputation consistency C_I , λ is set to 0.5. All models are trained using the Adam optimizer with a learning rate of 1×10^{-3} , L2 regularization weight of 5×10^{-4} , and hyperparameters β_1 and β_2 as 0.9 and 0.99, respectively. For imputation task, training iterates 300 times and the early stop strategy is used with patience of 10. For anomaly detection task, training iterates 8 times on stage 1 and 30 times on stage 2 with a patience of 5. The batch size for all experiments is set to 128.

D. Results and Discussion

1) *Imputation*: The imputation performance is reported in Table II, and the related visualization of the critical difference diagram based on the Wilcoxon pairwise signed-rank test after rejecting the null hypothesis using the Friedman test of the metric 1/mse is shown in Fig. 6. Compared with all baselines, the MoPIN achieves the best performance. Specifically, the MoPIN outperforms the SOTA baseline by relative 22.17% ($0.083 \rightarrow 0.065$), 8.05% ($0.189 \rightarrow 0.174$), and 11.81% ($0.288 \rightarrow 0.254$) on the mse, the MAE, and the RMSE on the ETTh1. And on the Air, it outperforms the SOTA baseline by relative 23.21% ($0.321 \rightarrow 0.247$), 30.36% ($0.228 \rightarrow 0.159$), and 19.44% ($0.501 \rightarrow 0.404$) on the mse, the MAE, and the RMSE.

It is worth noting that since deep learning-based methods consider more information in series modeling, all deep learning-based methods outperform traditional methods on

TABLE II
IMPUTATION PERFORMANCE

Dataset	ETTh1			Air		
Method	MSE	MAE	RMSE	MSE	MAE	RMSE
Mean	1.047	0.762	1.023	1.206	0.765	1.068
MissingForest	1.065	0.764	1.032	1.216	0.764	1.070
SoftImpute	0.551	0.515	0.742	0.559	0.398	0.703
MatrixFactorization	0.582	0.529	0.763	0.553	0.429	0.703
AMDAE	0.332	0.415	0.575	0.506	0.390	0.662
BRITS	0.164	0.271	0.405	0.347	0.228	0.514
MRNN	0.335	0.425	0.578	0.368	0.312	0.606
AIA-GRU	0.269	0.358	0.519	0.501	0.394	0.659
SPIN	0.260	0.354	0.510	0.326	0.250	0.504
TimesNet	0.083	0.189	0.288	0.321	0.271	0.501
PatchTST	0.181	0.297	0.425	0.374	0.292	0.550
SAITS	0.127	0.255	0.356	0.371	0.320	0.594
MoPIN	0.065	0.174	0.254	0.247	0.159	0.404

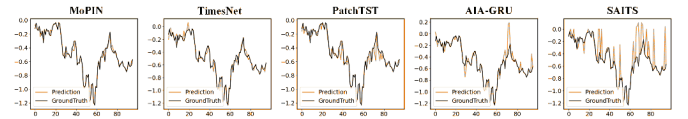


Fig. 7. Imputation result visualization on ETTh1.

both benchmarks. In addition, it is found that the performance of the SoftImpute and the MatrixFactorization methods far exceeds the Mean and the MissingForest, which is because these two methods gradually impute in an iterative manner and achieve higher accuracy in the process of gradual correction. This also provides indirect motivation support for the two-step imputation design of the MoPIN. However, this iterative approach brings a huge amount of time consumption while achieving high precision. Therefore, the MoPIN improves the performance by only two-step imputation without multiple iterations, making it advantageous in both accuracy and time consumption.

Meanwhile, among the deep learning-based methods, except for the PatchTST and the TimesNet, which are time-series analysis methods, the rest are ad hoc methods specially designed for imputation. However, except the SAITS, the other methods cannot outperform the PatchTST and the TimesNet. From the perspective of method design, this can be attributed to the powerful ability of series information extraction based on the patch. In addition, the SAITS significantly outperforms the PatchTST on ETTh1 and has similar performance on Air, which is because unlike other deep learning-based methods, the SAITS is also designed with a multistep imputation approach, which provides it with excellent performance.

In addition, for the MoPIN, compared with the TimesNet and the PatchTST, the design of multiscale PIMM of the MoPIN makes it more advantageous in extracting series information. And combined with the MoPIN proprietary imputation design, it achieves the best performance on both benchmarks. In order to clearly demonstrate the performance of multiple methods, Fig. 7 visualizes the results of the MoPIN and the other four best baselines on ETTh1.

Furthermore, the imputation performance of the MoPIN under different missing rates is verified on ETTh1. Specifically, the imputation results of the MoPIN are compared with several

¹Registered trademark.

²Registered trademark.

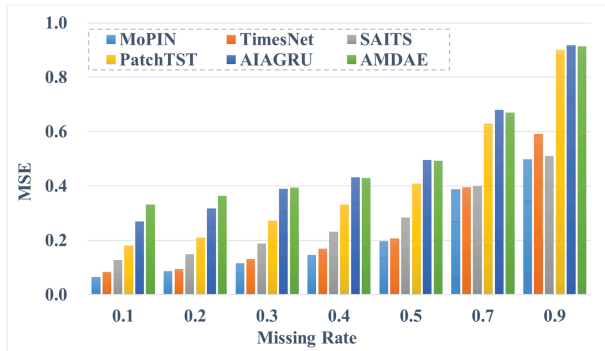
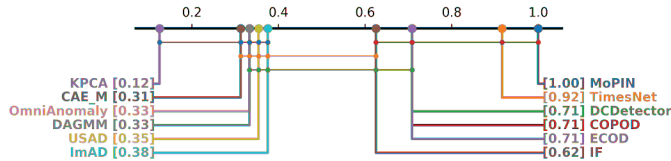


Fig. 8. Imputation performance under different missing rates on ETTh1.

Fig. 9. Critical difference diagram of the metric $F1$.TABLE III
ANOMALY DETECTION PERFORMANCE (%)

Dataset	PSM			Spacecraft		
Method	P	R	F1	P	R	F1
KPCA	71.15	89.99	79.46	52.07	100.0	61.97
IF	95.25	83.88	89.21	87.53	85.69	86.60
COPOD	95.55	84.44	89.65	92.06	85.69	87.63
ECOD	89.58	86.88	88.21	93.71	85.69	88.56
DAGMM	81.48	82.47	81.97	72.30	100.0	83.76
OmniAnomaly	84.67	81.59	83.10	68.93	100.0	81.58
USAD	84.47	81.59	83.01	71.58	100.0	83.31
CAE_M	85.95	80.26	83.01	70.44	100.0	82.61
DCDetector	87.09	89.33	88.19	81.05	98.83	89.01
ImAD	86.30	90.30	88.25	62.74	100.0	74.35
TimesNet	99.37	86.44	92.45	80.91	100.0	89.45
MoPIN	97.70	94.20	95.91	88.76	100.0	93.93

baselines at missing rates of 0.1–0.9 by random masking. The experimental results are shown in Fig. 8. It can be seen that when the missing rate is between 0.1 and 0.4, the MoPIN has a significant advantage. When the missing rate is between 0.5 and 0.9, the MoPIN still has a slight advantage over the best baseline. In general, the MoPIN can adapt to impute in low and extreme missing rate scenarios.

2) *Anomaly Detection*: The anomaly detection performance is reported in Table III, and the related visualization of the critical difference diagram of the metric $F1$ is shown in Fig. 9. Compared with all baselines, the MoPIN achieves the best performance. Specifically, the MoPIN outperforms the SOTA baseline by relative 3.74% (92.45 $\rightarrow$ 95.91) and 5.01% (89.45 $\rightarrow$ 93.93) on PSM and Spacecraft on $F1$ metric.

For anomaly detection in series with sparse observations, all baselines require prior imputation, which makes them highly dependent on the performance of the imputation model. Although the SOTA imputation baseline the TimesNet in experiments is used, this still hinders anomaly detection performance because the imputed series may introduce additional information different from the original series. This

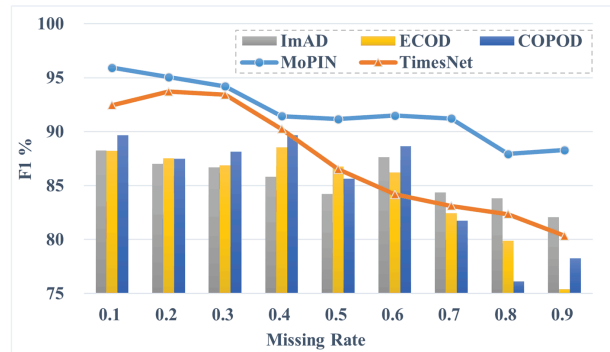


Fig. 10. Anomaly detection performance under different missing rates on PSM.

extra information is intolerable for the anomaly detection task, as it goes against the core idea of USAD. Therefore, the SOTA baseline DCDetector on complete series cannot achieve good anomaly detection on a sparse series. Although the baseline ImAD avoids preimputation problem, it does not fully use the information containing in data, while the MoPIN can achieve better performance because of its mixture of information modeling approach. In addition, the TimesNet is the best-performing baseline. On the one hand, this is due to the patch-based method design. On the other hand, the TimesNet is also used as the imputation method before anomaly detection and has the same model structure in both tasks, which gives it an advantage in task transfer. However, it is still a two-stage anomaly detection method. The MoPIN uses imputation consistency to avoid the gap between sparse series and anomaly detection, which is the core of achieving the best anomaly detection performance.

Furthermore, the anomaly detection performance of the MoPIN under different missing rates is verified on PSM. Specifically, the anomaly detection results of the MoPIN are compared with several baselines at miss rates ranging from 0.1 to 0.9 by random masking. The experimental results are shown in Fig. 10. From the results, it can be seen that although the TimesNet has a performance close to the MoPIN at a low missing rate, the performance of the TimesNet decreases rapidly as the missing rate increases. For the ImAD, it does not need a preimputation step, so it performs better on high missing rate compared with other baselines, which further shows the advantage of a single-step detection strategy for sparse series. Overall, the TimesNet is the SOTA baseline, benefiting from its unified model for imputation and anomaly detection. However, the performance degradation at high missing rate shows that the two-stage anomaly detection is not beneficial for sparse series. The MoPIN well unifies the relationship between anomaly detection and imputation, can adapt to anomaly detection task in low and extreme missing rate environments, and has better universality than baselines.

Overall, the above experimental results and analysis demonstrate the effectiveness of the MoPIN in time-series imputation and anomaly detection.

E. Ablation Study

To verify the motivation of the MoPIN, ablation experiments are conducted on the three main components of the MoPIN. And the results are shown in Table IV.

TABLE IV
ABLATION RESULTS

Dataset	ETTh1 (Imputation)			PSM (Anomaly Detection)		
Model	MSE	MAE	RMSE	P	R	F1
MoPIN	0.065	0.174	0.254	97.70	94.20	95.91
w/o C-I	0.103	0.226	0.320	94.23	93.61	93.92
w/o C-M	0.115	0.235	0.339	97.62	90.52	93.94
w/o A-A	0.074	0.191	0.271	94.45	91.95	93.18
w/o M-S	0.073	0.190	0.270	92.45	95.92	94.16
w/o T-S	0.093	0.216	0.305	94.15	90.52	92.30
w/o D-C	-	-	-	97.53	91.40	94.37
w/o I-C	-	-	-	94.94	93.05	93.99

1) *PIMM*: The effectiveness of the PIMM is verified through four variants of the MoPIN: 1) the MoPIN without channel independence encoding (w/o C-I); 2) the MoPIN without channel mixture encoding (w/o C-M); 3) the MoPIN without adaptive aggregating (w/o A-A); and 4) the MoPIN without multiscale modeling (w/o M-S). The experimental results show that in both tasks, the performance of the variants w/o C-I and w/o C-M has a significant decrease, which shows that both channel independence and mixture modeling play an important role in the MoPIN. In addition, compared with the MoPIN, the performance of w/o A-A and w/o M-S has declined to a certain extent, which shows that multiscale modeling contributes to the MoPIN. Therefore, the ablation experiments of the above four variants verify the role of the PIMM module in the MoPIN.

2) *Two-Step Imputation*: From the comparative experimental analysis, it can be seen that iterative imputation contributes to the imputation performance. Therefore, the MoPIN without two-step imputation (w/o T-S) is used to verify its effectiveness on imputation and anomaly detection. From the experimental results, it can be seen that the two-step imputation contributes to the performance of the MoPIN (0.065 $\rightarrow$ 0.093 on mse and 95.91% $\rightarrow$ 92.30% on $F1$). In particular, since the two-step imputation can achieve better imputation performance, it enables it to have better anomaly detection performance for the MoPIN, which detects anomalies using imputation consistency.

3) *Imputation Consistency*: Imputation consistency consists of two parts: the mse and distribution similarity of two imputed series. The effectiveness of imputation consistency is verified through two variants of the MoPIN: 1) the MoPIN without distribution consistency (w/o D-C) uses imputation error between two imputed series to detect anomalies and 2) the MoPIN without imputation consistency (w/o I-C) uses reconstruction error between the reconstructed series and original series to conduct anomaly detection, which is the same as baselines. From the experimental results, without using imputation consistency, the $F1$ metric of the variant w/o I-C using the two-stage anomaly detection strategy drops from 95.91% to 93.99% compared with the MoPIN, which validates the role of imputation consistency in anomaly detection for time series with sparse observations. In addition, relative to the MoPIN, the $F1$ of the variant w/o D-C drops from 95.91% to 94.37%, which shows the importance of adding a distribution consistency constraint.

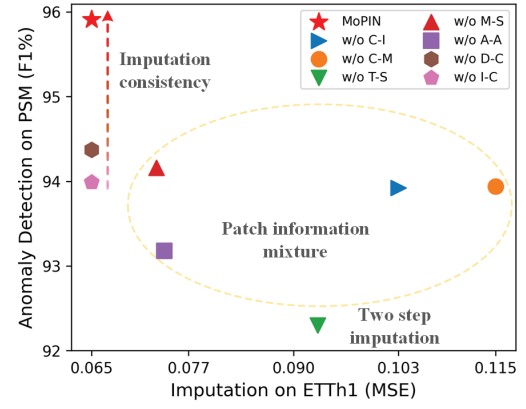


Fig. 11. Ablation analysis in anomaly detection (↑) and imputation (←). For variants w/o D-C and w/o I-C, the mse is set to be the same as the MoPIN.

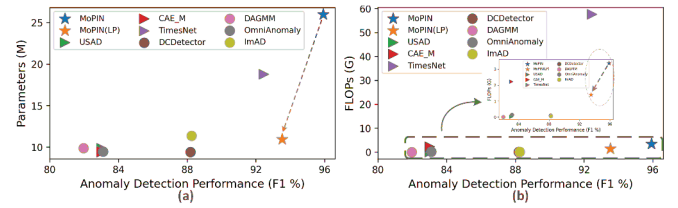


Fig. 12. Relationships between the anomaly detection performance and model efficiency on (a) parameters and (b) FLOPs.

For intuitiveness, Fig. 11 summarizes the performances of the above variants in two tasks, which shows that every component contributes to the MoPIN. In summary, the above results and analysis prove the correctness of the motivation of the MoPIN.

F. Efficiency Analysis

The number of parameters and the floating point operations (FLOPs) [36] are used to evaluate the memory and computational cost of the MoPIN. Specifically, the above metrics are calculated using the experimental settings on the PSM dataset detailed in Sections IV-A and IV-B, and all parameters in the anomaly detection process are counted. The results are shown in Fig. 12. A higher $F1$ -score means a better anomaly detection performance, a lower FLOPs means the less amount of computation required by the model with better computation efficiency, and fewer parameters means better memory cost.

1) *Memory Cost*: From Fig. 12(a), the MoPIN achieves the best anomaly detection performance, but has a high parameter amount with 25.98M. However, in practical applications, the memory cost of the MoPIN is still scarce since monitoring algorithms are usually deployed on servers. In addition, for fair comparison, the number of parameters of the MoPIN is adjusted to keep basically consistent with the baselines. Specifically, the embedding and encoder dimensions d and h are set to 3 and 4. Under this condition, the MoPIN with 10.95M parameters achieves an anomaly detection performance on $F1$ of 93.53%, which means that the performance of the MoPIN has a significant advantage under the condition of the same memory cost.

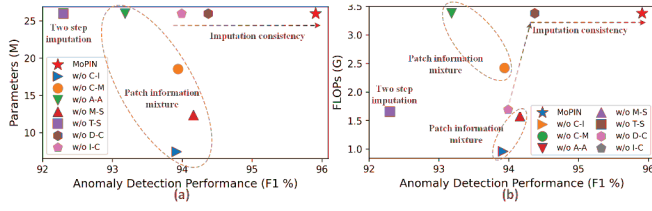


Fig. 13. Relationships between the anomaly detection performance and efficiency of variants of the MoPIN on (a) parameters and (b) FLOPs.

2) *Computation Cost*: From Fig. 12(b), the MoPIN and the TimesNet have relatively high FLOPs while achieving significant detection performance. However, for the actual training process, they can complete the training process in 3 min, which is little time consumption for offline modeling and online detection methods. Similarly, after reducing the number of parameters, the computational efficiency of the MoPIN has also been significantly improved while maintaining well anomaly detection performance. In addition, all models can predict a single sample in less than 1 ms, which is fully sufficient for actual industrial applications.

In summary, the MoPIN can achieve a well anomaly detection performance while maintaining the same memory consumption as the baselines. Moreover, by reducing certain efficiency conditions, the MoPIN can achieve better performance. In actual application scenarios, detection accuracy is usually more important, so tradeoffs can be made based on the performance of the monitoring server.

3) *Component Overhead*: In addition, in order to deeply analyze the source of the overhead of the MoPIN, the overhead contribution of each component in the ablation study is analyzed, and the results are shown in Fig. 13. Intuitively, the main memory overhead source of the MoPIN is from the PIMM, especially the channel independence encoding and multiscale modeling components. And the imputation consistency can significantly improve anomaly detection performance without requiring additional memory overhead. For computation overhead, almost all components contribute, and the most significant computation overhead still comes from channel independence encoding and multiscale modeling. In addition, both imputation consistency and two-step imputation strategies do not require memory overhead, but both have relatively large computation overhead. However, their improvement in anomaly detection performance is very significant. In practical applications, the parameters of each component can be traded based on the above analysis to achieve a balance between efficiency and detection performance.

V. VALIDATION ON A REAL-WORLD SCENARIO

The fluid catalytic cracking (FCC) process is one of the most important conversion processes in petroleum refining and is widely used to produce gasoline, diesel, olefin gas, and other products. In actual industrial applications, it is of great significance to ensure the safe and stable operation of the FCC. Considering the principle and complexity of the FCC, actual monitoring of the FCC process is often carried out on the three systems that constitute it: reaction regeneration, absorption stabilization, and fractionation system. Among them, the

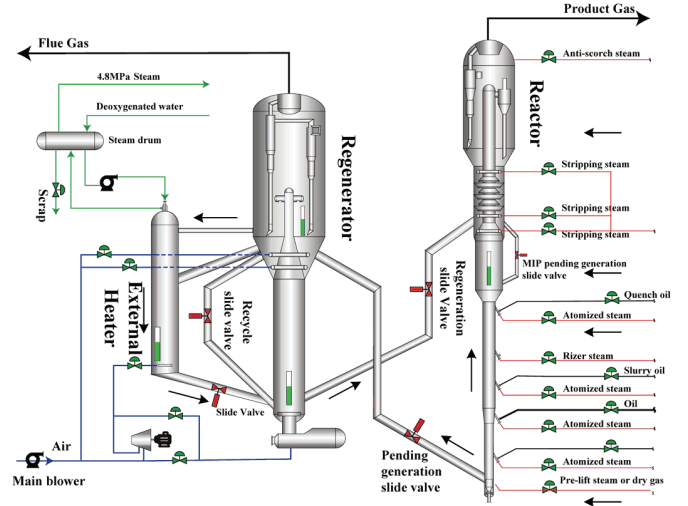


Fig. 14. Simplified diagram of the RRS.

TABLE V
KEY VARIABLES OF THE RRS

Variable	Description
v1	reactor top pressure
v2	regenerator top pressure
v3	external heater ladle pressure
v4	circulating slide valve pressure drop
v5	pending generation slide valve level
v6	recycle slide valve level
v7	reactor material level
v8	regenerator material level
v9	lift pipe outlet temperature
v10	reactor collector chamber temperature
v11	coke tank middle temperature
v12	regenerator bottom temperature
v13	regenerator outlet flue gas temperature
v14	main blower flow

reaction regeneration system (RRS) including the reaction and regeneration process is the core of the FCC. A simplified diagram of the RRS is shown in Fig. 14. In this section, a case of the RRS of the FCC in a petrochemical plant is provided for verification of the MoPIN.

According to the RRS process of the plant, Table V lists 14 key variables that can effectively reflect the operating status of the RRS. And the status monitoring of the RRS is implemented based on these 14 variables. The experimental dataset FCC-RRS with an average missing rate 12.8% was collected from the distribution control system of the petrochemical plant with a sampling period of 3 min from May to August in 2019. Two tasks are conducted on the FCC-RRS. For imputation, 29811 consecutive sampled series without anomalies are split into training, validation, and test sets with 15655/5155/8905 and 10% random masks on observations are applied as label data for evaluation. For anomaly detection, 2589 consecutive sampled series without anomalies are used for modeling and are split into training and validation sets with 1869/624, 24882 consecutive sampled series containing anomalies are used to test, and the anomaly ratio is 2.03%.

TABLE VI
IMPUTATION AND ANOMALY DETECTION
PERFORMANCE ON THE FCC-RRS

Task	Imputation			Task	Anomaly detection		
Method	MSE	MAE	RMSE	Method	P	R	F1
SoftImpute	0.973	0.576	0.965	IF	80.33	26.06	39.36
Matrix-Factorization	0.465	0.464	0.670	COPOD	84.54	93.09	88.61
AMDAE	1.285	0.788	1.108	ECOD	84.62	93.62	88.89
BRITS	1.052	0.578	1.002	DAGMM	79.66	100.0	88.68
MRNN	2.057	0.972	1.405	OmniAnomaly	85.92	99.46	92.19
AIA-GRU	1.437	0.807	1.171	USAD	89.95	90.95	90.45
SPIN	0.708	0.496	0.797	CAE_M	94.35	90.76	92.52
PatchTST	0.351	0.388	0.585	DCDetector	22.76	50.00	31.28
SAITS	0.807	0.554	0.841	ImAD	86.18	<u>99.47</u>	92.35
TimesNet	<u>0.266</u>	<u>0.351</u>	<u>0.511</u>	TimesNet	<u>96.59</u>	<u>94.65</u>	<u>95.61</u>
MoPIN	0.238	0.307	0.482	MoPIN	97.91	100.0	98.94

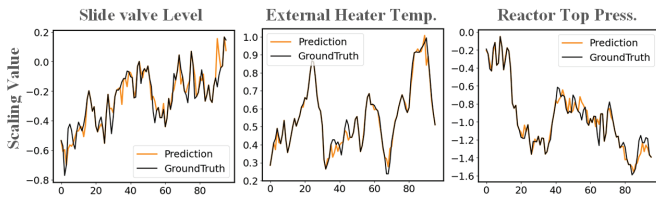


Fig. 15. Imputation visualization of the MoPIN on FCC-RRS.

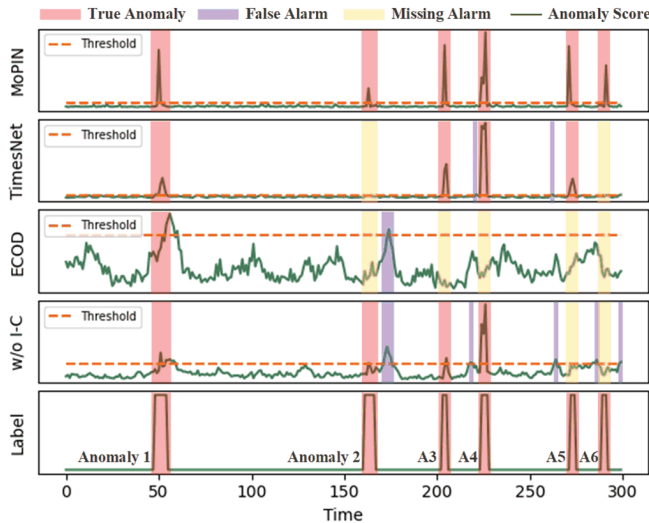


Fig. 16. Anomaly score comparison visualization on FCC-RRS.

The experimental result is shown in Table VI. For both tasks, the MoPIN achieves the SOTA performance compared with baselines. Specifically, the MoPIN outperforms the SOTA baseline by relative 10.38% ($0.266 \rightarrow 0.238$), 12.43% ($0.351 \rightarrow 0.307$), and 5.97% ($0.511 \rightarrow 0.482$) on the mse, the MAE, and the RMSE. And for the anomaly detection task, it outperforms the SOTA baseline by a relative 3.49% ($95.61 \rightarrow 98.94$) on $F1$. Moreover, compared with the TimesNet and the DAGMM, the MoPIN can detect all anomalies while ensuring a low false alarm rate, which is crucial for an actual industrial process.

In addition, for intuitiveness, Fig. 15 visualizes the imputation results of three key variables. It can be seen that the

MoPIN can impute missing values well, which is beneficial for further anomaly detection. And Fig. 16 shows the anomaly detection comparison performance of several methods by visualizing their anomaly scores. It can be seen that for most anomalies, the MoPIN has the largest anomaly score difference between normal and anomaly data, which can ensure both a high anomaly detection rate and low false alarm rate. In addition, it can be found that this advantage is weakened when not using imputation consistency, which further demonstrates its advantage.

In summary, the experimental results and analysis on the actual FCC process verify the effectiveness of the MoPIN and show that it has practical significance in ensuring industrial safety and avoiding economic losses.

VI. CONCLUSION

This article proposes a novel anomaly detection method for sparsely observed time series. Patch information mixture modeling is suggested to fully exploit the information of time series for accurate imputation and anomaly detection. In addition, a novel imputation consistency is proposed to achieve a single-stage anomaly detection directly from sparse series to anomaly scores. Extensive experiments on four real-world benchmarks in different domains demonstrate the effectiveness of the proposed method. Compared with baselines, the SOTA performance is achieved in both imputation and anomaly detection tasks. In addition, a real chemical plant FCC process application case is provided to verify the practical application significance of the proposed method. Specifically, the MoPIN achieved at least 8.05% mse relative improvement in imputation and 3.74% $F1$ relative improvement in anomaly detection.

Furthermore, although the performance of the MoPIN has been verified by relatively comprehensive experiments, it is still lack of theory and cannot provide a good interpretation for detected anomalies, which is also the same as current many famous works. Therefore, in the future, we will concentrate on theoretical analysis of the MoPIN and research on interpretable anomaly detection method for industrial time series. In addition, the MoPIN has relatively high computation cost, but if reducing parameters to keep basically consistence with baselines, it can still achieve better performance. Meanwhile, how to tradeoff the efficiency and performance in a real application is a practical problem. And refer to current works, a well tradeoff strategy can be made based on hardware performance and experiences.

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